Career advising:

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Call notes
Action items
    For advising team
    For Athreya
        (After the call) Define next actions
Fill in Before the Call:
What do you think are the most pressing problems in the world?
What kinds of work are you most suited for?
What are your options?
What are your key uncertainties?

Call notes

During the call, the advisor will take notes on the discussion here. Feel free to do so too, or not as you prefer.

Paths to value:

1. Get AI services to the world's poorest people sooner
2. AI will get used lots of places – making it much more reliable

For more "fundamental" "moonshot" kinds of projects:

- Maybe https://www.ycombinator.com/nonprofits/?

Methodologically, let's google stuff?

- https://www.google.com/search?q=effective+altruism+forum+malaria+vaccine&oq=effective+altruism+forum+malaria+vaccine&gs_lcrp=EgZjaHJvbWUyBggAEEUYOTIHCAEQIRigATIHCAIQIRigATIHCAMQIRigATIHCAQQIRiPAtIBCDg4NzIqMGo5qAIAsAIB&sourceid=chrome&ie=UTF-8
- https://www.givewell.org/international/technical/programs/disease-eradication
- https://blog.givewell.org/2024/09/10/research-strategy-vaccines/
- https://www.openphilanthropy.org/focus/

https://gatesfoundation.wd1.myworkdayjobs.com/en-US/Gates/job/Seattle-WA/Senior-Software-Engineer--Rapid-Reuse-Modeling_B020884-1
https://docs.google.com/document/d/12xLeRHtH9L9T2ht_f-BmkHiEh9ru4eVnz-HqfpJ1qro/edit?tab=t.0#heading=h.jt6jmxp623yt

Most theoretical

- https://www.alignment.org/theory/

DD networking strategy: treat it like a research project / literature review

- Have a goal of "knowing what everyone is doing in X space" (e.g. mirror life)
- "Wikipedia rabbit hole" + Keep a logbook; read lots of papers / blog posts / books / workshop proceedings, listen to lots of podcasts / watch videos of talks
- **When you find a project / team you think is cool** s.t. you kind of wish you'd been part of it, **email people who worked on it:**
    - Let them know that you got a lot out of it; most people think nobody ever reads their work, so this is a great foot to start on; maybe a specific sentence demonstrating that you really did read it :)

    - Tell them you're exploring the space to understand all the work that's happening
    - Ask them if they'd be willing to have a 30min phone call (I like to give them a calendly link so they can just sign up for a time right away) **to answer some questions about the work –** why they did it that way, what they think is promising to do next on the problem, the story of how they got there
    - If you have a good time, let them know at the end: "I'm thinking about moving my career towards this problem – if you hear about any jobs you think might be a good fit, I'd love to hear about them!"
- Advantages of this method:
    - They know what to expect on the call – you're not going to emotionally blackmail them into trying to get a job, or give them a conspiracy theory, whatever
    - You're immediately talking about a topic of shared interest, not about "career capital" or whatever
    - Hopefully talking to future colleagues, before you actually look for a job
- Later, if say there are a few people in some city you've had good chats with, you can contact them + say "hey I might be in Berlin next month; not sure my schedule, but if I was, could I buy you coffee / drop by the office / sit at a guest desk for a day?" + if a few of them say yes, you can actually do the trip :)

Post-call resources (AI)

Matt Beard's post-election takes:
"How are things in DC policy lately?"

Set up alerts! https://jobs.80000hours.org/

Reading

(Skewed towards what I think is most accurate and useful! See the 80k website as a starting point for other reading)

Lesswrong summaries might not be exhaustive, but can get you started:

- Shallow review of technical AI safety, 2024 — LessWrong
- (My understanding of) What Everyone in Technical Alignment is Doing and Why — LessWrong
- What is everyone doing in AI governance — LessWrong

What's the big picture?

- What we're doing here
- What failure looks like
- The case for AI x-risk (me; I still roughly agree w/ this)
- Preventing an AI-related catastrophe (a coworker of mine at 80k)
- AI Could Defeat All Of Us Combined
- Carl Shulman (Part 1) on the 80,000 Hours podcast
- Some old alignment research I think is still useful for orientation:
    - AI safety via debate
    - Supervising strong learners by amplifying weak experts
    - 11 Proposals for Building Safe AI

I don't currently have great high-level links for the AI governance problem / potential solutions; hoping to fix this in the future.

What's the plan? Very broadly:

- Solve technical / engineering problems to make loss of control less likely
- Figure out what governance of powerful AI should look like, and help governments implement sensible policies today + lay the groundwork for what we'll need in the future

Not everyone thinks of it this way; groups like Pause AI and MIRI have pretty different-feeling plans.

Jobs that could help / rough map of organisations

Note: I don't think these orgs have good enough plans that you can just join and expect to be having a good impact; I think you need to evaluate the specific role, team, and projects and see whether you think it's having the kind of impact you want.

Some technical AI safety organisations (mostly nonprofit, alphabetical order):

- AI Alignment Research Center
- Apollo Research
- Center for AI Safety
- Elicit / Ought
- FAR AI
- METR
- Palisade Research
- Redwood Research
- UK AI safety institute
- US AI safety institute (at NIST)
- Subsection on information security:
    - Our career guide on information security

        - …especially how to assess your fit in advance
        -
    - Securing AI Model Weights: Preventing Theft and Misuse of Frontier Models | RAND
    - Palisade Research

Some government jobs:

- Legislative staffing: every legislator, national and state-level, has some staff helping them figure out what to do, and very few are well-advised about AI.
- US AI safety institute (at NIST)
- UK AI safety institute
- US OSTP
- (A ton of other jobs in many governments could go here)
- Horizon Institute: technically this is more of a step along the road than an endpoint, but it's important to know about.

Some nonprofits / think tanks:

- CSET
- Centre on Long Term Resilience
- Centre for the Governance of AI
- Institute for Law & AI
- Simon Institute
- Future Society
- RAND
- CAIP
- CAIS

Advocacy orgs:

- Pause AI
- MIRI

Companies

- Anthropic and DeepMind each have several teams, technical and policy, that I believe are working to avert AI x-risk. They're also pushing AI capabilities towards the point where those risks will come real. Great example of the complexity I mention above.

Other notes:

- OpenAI and other big AI companies: I'm not recommending that anyone go to these jobs right now; these orgs don't seem to be prioritising AI risk, and OpenAI in particular has track record such that I don't think you can trust them to deal with you fairly or make good decisions. But some other advisors disagree and are still recommending

security and/or safety research roles here to some people, so for completeness I wanted to mention them.
- See our career reviews here for possibilities we might not have discussed

More: the 80000 Hours job board, including the Organisations and Collections tabs.
- Conor, who runs the jobs board, says it works similarly to the note I gave above: listing a job on the board means we think it's worth investigating / considering seriously, not that we've already investigated deeply and are confident that the job is great, and not that the organisation as a whole is having a positive impact in the world.

Building skills

For policy / governance:
- Emerging Tech Policy Careers – this is a very good overview of basically how things work in DC, at least according to the authors; reading all of it would be time consuming, but I learned a ton just from my first shallow pass over it.
- Horizon Institute
- GovAI summer/winter fellowships – seems likely these will keep happening
- AI Safety Fundamentals, Governance Course - BlueDot Impact

Structured programs for ML and/or alignment:
- AI Safety Fundamentals – I highly recommend everyone interested in this area at least look at the material for both the alignment and governance courses!
- https://www.arena.education/
- fast.ai ML Course
- Google ML Course
- ML Safety Course

Mentorship programs, alphabetical:
- Astra Fellowship (now closed for the year, hopefully will run again)
- LASR Labs
- MATS Program
- Pivotal fellowship, hopefully will be back
- Potential PhD advisors: this list could be fairly out of date, but I think it's something to start with
- Internships / Residency programs – as above, I haven't vetted this recently
- Supervised Program for Alignment Research

Events / community:

- Events & Training – AISafety.com – I don't keep a super close eye on this list as it's updated, so don't take this as an endorsement re: everything on the list

- AI Safety Support, especially lots of links
- Arkose: additional advising

Funding / grant programs

Some potential funding sources for career transition or AI xrisk projects:

- Request for proposals: AI governance | Open Philanthropy
- OpenPhil career development and transition funding (CoI: I used to be an AI grantmaker there, so I'm not plausibly neutral)
- Request for Proposals: Technical AI Safety Research | Open Philanthropy
- Long-Term Future Fund
- Emergent Ventures | Mercatus Center
- CLR Fund
- Survival and Flourishing Fund, if you have an institutional home
- Founders Pledge
- Maybe: https://manifund.org/about/open-call
- A longer list: Funding opportunities - EA Forum
- Unvetted, but looks interesting: https://seldonaccelerator.com/
- Catalyze – AI Safety Incubation Program: maybe this will come back in the future?

Cheap compute for projects:

- Google Colab and Kaggle
- Amazon and Google have student compute credit programs
- New ones continue to appear; basically you should think there's probably a way to get pretty cheap compute, so if that's your bottleneck, have a look around.

Action items

For advising team

What the advisor plans to do next. These are to be added during the call by the advisor.

1.

For Athreya

Suggestions for what the advisee should do next. These are to be added during the call by both the advisor and the advisee.

1.

(After the call) Define next actions

Write out what you're going to do—in the form of small, manageable next actions, as well as and *when* you'll do them.

Next actions	**When you'll do them**
This week	
This month	
6 months	

And so on…

For anything you can't do right away, add a reminder in your calendar.

Keep adding to this list (right click to insert a row below), taking your next actions, and crossing items off as long as you find it useful.

Fill in Before the Call:

What do you think are the most pressing problems in the world?

What is your definition of positive impact?

We know this is a very big question. You don't have to have a full worked out view here. We're just interested to hear your own thoughts on this, answering this question however makes sense to you. If you're wondering, our thoughts are here.

Enabling more people to be free from suffering over the long term.
I like your definition. Though I am weighted much more towards current humans (as opposed to future humans, non-human animals, digital beings etc.). While I think the musings about wellbeing are valuable, I want to focus on the basics - water, food, shelter, safety, free from disease.

What global problems do you think are most pressing? Try to rank them as best you can. We've listed below the ones you ticked on the application form to provide inspiration but feel free to change your answers or write more.

1. Global poverty and health: this is by far the most important in my eyes.
2. Risks from general, agent-like artificial intelligence: I am not sure if I believe in xrisk due to AI. But I am increasingly starting to feel that AI has massive potential to make breakthroughs in Science which would speed up (1). But building such AI should be done safely.

What kinds of work are you most suited for?

What do you most enjoy? What jobs do you think you could become good at?

Our research focuses on which career paths and roles are most impactful if done well, so that's what we'll discuss most in our conversation. But you're much more likely to have a large impact if you find a role you're well suited to and will enjoy over the long term. So we encourage you to think hard about what

types of activities you're well and badly suited to, and which you'd like to try out. You might find our article on 'How to identify your personal strengths' useful for that investigation.

What I enjoy

I enjoy fundamental/clean problems. I like working on things where the challenges are innate to the problem.

E.g. I don't enjoy (generic) software engineering at big tech because the problems being solved aren't complicated in themselves, they are hard because of the way the codebase and tools were architected.

On the other extreme, I enjoyed theoretical CS research though I didn't have much success.

I also enjoy building and experimenting.

Strengths

Teaching/presenting: I seem to get good reviews from peers/students/colleagues when I, for example, taught algorithms, did seminars, or conducted trainings at work.

Programming, Data Structures and Algorithms: I did decently at sport programming in college and reached division 1 on Codeforces.

Discrete math/linear algebra: during my master's, I did end up doing a bunch of discrete math. Not good by any means but I probably have much more exposure than the average person.

Ability to execute: If given a clear direction, I am very good at executing independently.

Learn deeply: I don't learn quickly, but I do learn deeply.

People skills: Patience. Good active listener, decent at understanding other perspectives.

Wide interests: I enjoy learning about a range of things.

Good with languages: I seem to pick up the basics languages quickly.

What I might be suited for

I feel like I could succeed at research. I am not very "quick," but given time, I can pick up complicated things. However, I had a bad experience during my MS by Research which ended in me not pursuing a PhD.

Key uncertainties: I struggle following through on long-drawn efforts and get distracted by the "next cool thing". I also struggle when there is no clear indicator of progress.

I also feel like I can be some sort of 3Blue1Brown style content creator.

Key uncertainties: Now with LLMs becoming increasingly good personal tutors, I find it hard to see counterfactual impact.

What are your options?

Below are your plans as you described them on the application form. It's useful for us to have a bit more detail on them for our call.

Err on the side of listing too many than too few, and also consider longer term options. Briefly write about why they might and might not be the right option.

Stay in Software Engineering at Big Tech/build tech products, and E2G into effective charities.
Pro: Already have the career capital, no lifestyle changes required, very very relaxed.
Con: Does not directly work on impactful problems, I don't enjoy it/don't feel intellectually challenged.

Do intersectional research to contribute towards AI interpretability.
I don't know much about this field, but when reading about things from linguistics or cognitive science, I always wonder whether models work similarly. A lot of these feel like evaluation ideas rather than interpretability.
Pro: leverages my natural curiosity and interest in multiple fields.
Con: Unsure of impact, no credentials, don't want to go back to school full-time if I can avoid it.

Become a YouTube essayist/some sort of content creator creating explainer videos on EA related topics.
Pro: leverages my strong ability to teach,
Con: No direct impact, unsure of counterfactual impact as described in the last section.

Do AI alignment research.
Pro: Direct impact, leverages my interests and skills in math, algorithms, industry work available
Cons: No credentials, don't want to go to school full time if I can avoid it, unclear if most alignment research will be useful.

Data science/software for policy/bio safety/food safety.
Pro: work directly on what I think is most impactful (I'm bigger on these than existential risk due to AI), clear and assured impact.
Cons: unclear what opportunities are available, unclear what skills they require, unclear if I'll need to reskill.

I (possibly naively) feel that there is incredible potential at the intersection of CS/ML and Bio. Especially since AlphaFold and Isomorphic. However, I have no credentials in either Bio or ML

and don't know anything about things like bioinformatics and whether there is actual impact to be made there.

Tech Entrepreneurship/Indie Hacking and E2G.
Pro: independence, leverages existing software skills
Con: I don't think I have the risk appetite or business acumen to make this work

Quant/Algo trading and E2G
Pro: super lucrative, intellectually challenging
Con: I do terribly with excessive stress, too risk averse, bad at quick decision making, not a math olympiad medalist :P

Quantum Computing Research
Pro: intellectual fit, if it works will contribute to drug discovery and data generation for learning models and solve other problems
Cons: unclear timeline, unclear feasibility, unclear impact

Theory CS research
Pro: intellectual fit, career capital
Con: no clear impact, scope limited to academia, very difficult

What are your key uncertainties?

Information that you could go and get (like by speaking to someone), that could change your options. Under each one, write out how you could answer this key uncertainty (what small concrete actions could you take over the next month to answer questions / uncertainties?). We can chat about how to do this on the call if you don't have the answers.

1. Visa issues - I am Indian, so I will require visa sponsorship for everything. I am looking for roles primarily in the EU/UK. I don't see a realistic long-term path in the US.
2. AI risk - The rate of AI progress makes me wonder what jobs will remain relevant.
3. Breaking in - I don't have the credentials for people to give me a shot. I don't have letter writers to get me into a PhD.
4. Academia - I don't want to be in academia. I want to avoid doing a PhD if I can manage it. Open to changing my mind on both.
5. Research fit - I learnt the hard way how big a role the supervisor and research group play in doing good research. I find it scary that this is hard to judge without actually working with the person.
6. Research obsolescence - At the rate the field is changing, I fear problem statements will remain relevant for very little time.
7. Research value - Only the top percentile of research ends up being read/adds value.
8. Leaving a "dream" job - the effort to savings ratio is ridiculously good, and even though I don't enjoy it, it will make me financially secure if I continue.

9. Arrival fallacy - That after I find something that fits all my criteria I am still not satisfied.